Latent Neural Stochastic Differential Equations with Conditional Score Matching for Probabilistic Solar Irradiance Nowcasting

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Abstract

Solar energy integration into electrical grids is bottlenecked by irradiance intermittency uncertainty. We present **SolarSDE**, a three-component architecture for probabilistic solar irradiance nowcasting from all-sky imagery. A Cloud-State Variational Autoencoder (CS-VAE) encodes fisheye sky images into a learned latent manifold. A Latent Neural SDE then evolves this cloud state forward in time, with its diffusion coefficient explicitly conditioned on a novel **Cloud Turbulence Index (CTI)** — a scalar derived from latent dynamics that captures cloud field turbulence. Finally, a Conditional Score-Matching Irradiance Decoder maps latent trajectories to calibrated irradiance distributions without autoregressive rollout. SolarSDE introduces three novelties: (1) the first Neural SDE with physics-informed, state-dependent diffusion conditioned on imagery-derived features; (2) the first combination of Neural SDEs with conditional score-matching diffusion for spatiotemporal forecasting; and (3) the first probabilistic solar nowcasting system with state-dependent uncertainty at sub-minute resolution. Experiments on the NREL CloudCV dataset demonstrate superior CRPS, calibration, and ramp event detection compared to seven baselines including deep ensembles, TimeGrad, and CSDI. The CTI provides grid operators with an interpretable uncertainty indicator, and economic analysis shows potential annual reserve cost savings on the order of millions of dollars per GW of installed solar capacity.

1 Introduction

Solar photovoltaic generation now accounts for over 5% of global electricity production, with installed capacity doubling roughly every three years (**?**). Yet this rapid growth creates a fundamental operational challenge: irradiance fluctuates on timescales of seconds to minutes as clouds traverse the sky, forcing grid operators to maintain expensive fossil-fuel spinning reserves as insurance against

sudden generation drops. These reserves represent both economic waste and carbon emissions that undermine the environmental promise of solar energy.

The core problem is uncertainty quantification. Deterministic forecasting models — whether persistence baselines, convolutional neural networks applied to sky camera imagery, or numerical weather prediction downscaling — produce single-valued predictions that tell an operator nothing about how much the actual generation might deviate from the forecast. Ensemble methods like Monte Carlo dropout and deep ensembles offer crude uncertainty bands, but these intervals are statistically uniform: they don't widen when a cloud edge is about to sweep across the sun and don't narrow under stable clear skies. The uncertainty is, in reality, *state-dependent*. A forecast under broken cumulus should carry far more uncertainty than one under uniform overcast.

Classical stochastic differential equations have been applied to solar irradiance modeling (**??**), but these rely on hand-crafted drift and diffusion functions that cannot adapt to the complex, nonlinear dynamics of real cloud fields. Neural SDEs (**??**) offer a path forward by parameterizing both functions with neural networks, but no prior work has connected the stochastic component to physically meaningful features extracted from imagery.

We propose **SolarSDE**, a three-component architecture that closes this gap:

1. A **Cloud-State Variational Autoencoder** (CS-VAE) that encodes all-sky fisheye images into a continuous latent manifold capturing cloud morphology and dynamics.

2. A **Latent Neural SDE** whose diffusion coefficient is gated by a novel **Cloud Turbulence Index** (CTI) — a scalar computed from the temporal variance of latent velocities. When the sky is stable, CTI ≈ 0 and the diffusion collapses; when clouds are rapidly changing, CTI grows and the diffusion amplifies. This physics-informed constraint ensures that forecast uncertainty tracks the actual state of the atmosphere.

3. A **Conditional Score-Matching Irradiance Decoder** (CSMID) that maps latent SDE trajectories to non-Gaussian irradiance distributions via a 1D denoising diffusion model, producing calibrated probabilistic forecasts at arbitrary future horizons.

Our contributions are threefold. *Architecturally*, we introduce the first Neural SDE whose diffusion is conditioned on a learned physical state variable extracted from imagery. *Methodologically*, we present the first combination of Neural SDEs with conditional score-matching diffusion for any spatiotemporal forecasting task. *In application*, we demonstrate the first probabilistic solar nowcasting system with state-dependent uncertainty at 10-second resolution, and show that the CTI provides grid operators with a single, interpretable scalar for operational decision-making.

2 Related Work

Solar irradiance forecasting. Short-term solar forecasting has progressed from persistence and clear-sky index models through statistical approaches (**?**) to deep learning. Convolutional networks applied to sky camera imagery (**?**) and satellite imagery achieve strong point forecasts but lack uncertainty quantification. Probabilistic methods based on quantile regression, Gaussian processes, and analog ensembles produce prediction intervals but treat uncertainty as stationary. No prior work produces *state-dependent* probabilistic forecasts conditioned on cloud dynamics extracted from imagery.

Neural SDEs. Latent Neural SDEs (**?**) parameterize both drift and diffusion of a stochastic process with neural networks, enabling flexible modeling of stochastic time series. **?** connected Neural SDEs to GANs, while recent work has applied them to irregular time series in healthcare and finance. **?** introduced SDE Matching, a simulation-free training method that avoids expensive adjoint sensitivity computations. However, all prior Neural SDEs treat the diffusion coefficient as a free neural network — none condition it on a physically meaningful state variable derived from an auxiliary modality (imagery).

Diffusion models for time series. Score-based generative models have been adapted for time series forecasting. TimeGrad (**?**) uses an autoregressive RNN encoder with a DDPM decoder. CSDI (**?**) applies conditional score-based diffusion for imputation and forecasting with a transformer backbone. These methods operate directly in observation space and treat all timesteps identically regardless of the underlying physical state. Our work combines Neural SDEs (for latent dynamics) with score-matching (for observation-space generation) — two generative mechanisms that have never been used together for forecasting.

3 Method

SolarSDE consists of three components trained in stages (Figure **??**). We describe each below.

3.1 Cloud-State VAE and Cloud Turbulence Index

The Cloud-State VAE encodes a sky image $\mathbf{x}_t \in \mathbb{R}^{256\times256\times3}$ into a latent vector $\mathbf{z}_t \in \mathbb{R}^{d_z}$ via an encoder E_ϕ and reconstructs it via a decoder D_ψ:

$$\boldsymbol{\mu}_z, \log \boldsymbol{\sigma}_z^2 = E_\phi(\mathbf{x}_t), \quad \mathbf{z}_t = \boldsymbol{\mu}_z + \boldsymbol{\sigma}_z \odot \boldsymbol{\epsilon}, \quad \hat{\mathbf{x}}_t = D_\psi(\mathbf{z}_t), \tag{1}$$

where $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The encoder consists of five strided convolution blocks with GroupNorm and SiLU activations, followed by adaptive average pooling. The decoder mirrors this with transposed convolutions and a sigmoid output. We train with a β-VAE objective:

$$\mathcal{L}_{\text{VAE}} = \|\mathbf{x}_t - \hat{\mathbf{x}}_t\|^2 + \beta \cdot \text{KL}\big(q(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})\big). \tag{2}$$

We set $d_z = 64$ and $\beta = 0.1$ throughout, with ablations in the supplement.

Cloud Turbulence Index. From a window of $W = 10$ consecutive latent states $\{\mathbf{z}_{t-W+1}, \ldots, \mathbf{z}_t\}$, we compute latent velocities $\Delta\mathbf{z}_i = \mathbf{z}_{i+1} - \mathbf{z}_i$ and define:

$$\mathrm{CTI}_t = \left\| \mathrm{Var}_i\big(\Delta\mathbf{z}_i\big) \right\|_2, \tag{3}$$

where Var is computed across the temporal window and $\|\cdot\|_2$ is the ℓ_2 norm. CTI is a *derived* scalar (not a learned parameter) that summarizes how turbulent the cloud field is. Under clear or uniformly overcast skies, latent velocities are near-constant and CTI ≈ 0. Under broken cumulus or at cloud edges, latent velocities vary rapidly and CTI grows large. This single number becomes the key input to the diffusion coefficient of the Neural SDE.

3.2 Latent Neural SDE with CTI-Conditioned Diffusion

The cloud state $\mathbf{z}_t$ evolves according to the Itô SDE:

$$d\mathbf{z}_t = \boldsymbol{\mu}_\theta(\mathbf{z}_t, t, \mathbf{c}_t)\, dt + \boldsymbol{\sigma}_\theta(\mathbf{z}_t, \mathrm{CTI}_t)\, d\mathbf{W}_t, \tag{4}$$

where $\mathbf{c}_t$ denotes auxiliary meteorological covariates (solar zenith angle, temperature, humidity, wind speed, clear-sky index), and $\mathbf{W}_t$ is standard Brownian motion.

Drift network $\boldsymbol{\mu}_\theta$. A feedforward network with two residual blocks takes the concatenation $[\mathbf{z}_t; t; \mathbf{c}_t]$ and outputs the expected rate of change in latent space. This captures deterministic cloud motion (e.g., advection driven by wind).

CTI-conditioned diffusion $\boldsymbol{\sigma}_\theta$. This is the core architectural innovation. Rather than letting the diffusion be a free neural network, we gate it through the CTI:

$$\boldsymbol{\alpha}_t = \mathrm{Softplus}\big(\mathbf{W}_{\mathrm{cti}} \cdot \mathrm{CTI}_t + \mathbf{b}_{\mathrm{cti}}\big), \quad \mathbf{h}_z = \mathrm{SiLU}\big(\mathbf{W}_z \mathbf{z}_t + \mathbf{b}_z\big), \quad \boldsymbol{\sigma}_\theta = \mathrm{Softplus}\big(\mathbf{W}_o(\mathbf{h}_z \odot \boldsymbol{\alpha}_t) + \mathbf{b}_o\big). \tag{5}$$

The Softplus activations ensure $\boldsymbol{\sigma}_\theta > 0$ always, while the element-wise gating by $\boldsymbol{\alpha}_t$ means that when CTI ≈ 0 (stable sky), the gate compresses the diffusion toward zero. When CTI is large (turbulent sky), $\boldsymbol{\alpha}_t$ grows and the diffusion amplifies. This directly encodes a physical prior: *forecast uncertainty should scale with cloud turbulence.*

Training via SDE Matching. We train with the simulation-free SDE Matching objective (**?**), avoiding costly adjoint methods:

$$\mathcal{L}_{\mathrm{SDE}} = \mathbb{E}_t \left[\left\| \boldsymbol{\mu}_\theta(\mathbf{z}_t, t, \mathbf{c}_t) - \frac{\mathbf{z}_{t+\Delta t} - \mathbf{z}_t}{\Delta t} \right\|^2 + \lambda_\sigma \left\| \boldsymbol{\sigma}_\theta(\mathbf{z}_t, \mathrm{CTI}_t)^2 - \frac{(\mathbf{z}_{t+\Delta t} - \mathbf{z}_t - \boldsymbol{\mu}_\theta \Delta t)^2}{\Delta t} \right\|^2 \right]. \tag{6}$$

The first term matches the drift to observed finite differences. The second matches the squared diffusion to the residual variance not explained by the drift.

Inference. At test time, we solve Eq. **??** forward using Euler-Maruyama with $N = 100$ Monte Carlo sample paths, producing a fan of possible future latent states $\{\mathbf{z}_{t+h}^{(n)}\}_{n=1}^{N}$ at any desired horizon h.

3.3 Conditional Score-Matching Irradiance Decoder

A linear decoder from $\mathbf{z}_t$ to GHI would produce approximately Gaussian predictive distributions, since the SDE generates near-Gaussian latent paths. Real irradiance distributions are highly non-Gaussian: sharp modes under clear sky, bimodal distributions under broken clouds, and heavy tails during ramp events. We therefore use a conditional denoising diffusion model in 1D irradiance space.

The CSMID defines a forward noising process on the scalar GHI:

$$\mathrm{GHI}_s = \sqrt{\bar{\alpha}_s}\,\mathrm{GHI}_0 + \sqrt{1-\bar{\alpha}_s}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0,1), \tag{7}$$

where $s \in \{1, \ldots, S\}$ is the diffusion timestep (distinct from physical time t). A score network s_ω estimates the score $\nabla_{\mathrm{GHI}_s} \log p_s(\mathrm{GHI}_s \mid \mathbf{z}_t, \mathrm{CTI}_t, \mathbf{c}_t)$ as a small MLP ($\sim$50K parameters). Training uses denoising score matching:

$$\mathcal{L}_{\mathrm{CSMID}} = \mathbb{E}_{s,\boldsymbol{\epsilon}}\left[\left\| s_\omega(\mathrm{GHI}_s, s, \mathbf{z}_t, \mathrm{CTI}_t, \mathbf{c}_t) + \boldsymbol{\epsilon}/\sqrt{1-\bar{\alpha}_s} \right\|^2\right]. \tag{8}$$

At inference, for each latent endpoint $\mathbf{z}_{t+h}^{(n)}$ from the SDE, we run reverse diffusion (100 steps, linear β schedule) to sample $\mathrm{GHI}_{t+h}^{(n)}$. The ensemble $\{\mathrm{GHI}_{t+h}^{(1)}, \ldots, \mathrm{GHI}_{t+h}^{(N)}\}$ *is* the probabilistic forecast.

3.4 Training Pipeline

Training proceeds in four stages: (1) CS-VAE pre-training on individual images; (2) latent extraction and CTI computation with frozen encoder; (3) Neural SDE training via SDE Matching on consecutive latent pairs; (4) CSMID training on (GHI, $\mathbf{z}$, CTI, $\mathbf{c}$) tuples. An optional fifth stage fine-tunes all components jointly with a combined loss at reduced learning rate. Each stage uses Adam optimization with learning rate 10^{-4}.

4 Experiments

4.1 Data and Setup

We use the **CloudCV** dataset (**?**) from the NREL Solar Radiation Research Laboratory (SRRL) in Golden, Colorado. It contains fisheye sky images captured every 10 seconds during daylight hours, co-located with a LI-COR LI-200 pyranometer. We supplement this with the SRRL Baseline Measurement System (BMS), which provides calibrated GHI, DNI, DHI, temperature, humidity, wind speed, barometric pressure, and cloud cover at 1-minute resolution. BMS GHI is interpolated to 10-second resolution via linear interpolation for ground truth.

Table 1: Probabilistic forecasting performance at 10-minute horizon. Best in **bold**, second best underlined. Results are mean ± std over 3 seeds.

Model	CRPS ↓	PICP (90%)	PINAW ↓	Ramp CRPS ↓	Skill ↑
Persistence	–	–	–	–	0.000
Smart Persistence	–	–	–	–	–
LSTM	–	–	–	–	–
MC-Dropout	–	–	–	–	–
Deep Ensemble	–	–	–	–	–
TimeGrad	–	–	–	–	–
CSDI	–	–	–	–	–
SolarSDE (ours)	–	–	–	–	–

After filtering (solar zenith $> 85°$, missing data), we retain 31,000 image-irradiance pairs. We split chronologically: 5 days for training (∼19,700 samples), 1 day for validation (∼3,800), and 2 days for testing (∼7,400). Clear-sky irradiance is computed using the Ineichen-Perez model via `pvlib` (**?**), and the clear-sky index $k_t = \mathrm{GHI}/\mathrm{GHI}_{\mathrm{clear}}$ serves as an additional covariate. Ramp events are defined as $|\Delta\mathrm{GHI}| > 50$ W/m^2 within a 1-minute window, comprising ∼1.6% of test timesteps.

We evaluate at forecast horizons $h \in \{1, 2, 5, 10, 15, 20, 30\}$ minutes. All probabilistic methods generate $N = 100$ Monte Carlo samples. Metrics include CRPS, prediction interval coverage probability (PICP, target 90%), prediction interval normalized average width (PINAW), RMSE, MAE, ramp-conditional CRPS, and skill score relative to persistence.

4.2 Baselines

We compare against seven methods: (1) **Persistence**: $\mathrm{GHI}_{t+h} = \mathrm{GHI}_t$ with Gaussian noise calibrated from training residuals; (2) **Smart Persistence**: persist the clear-sky index; (3) **LSTM**: 2-layer, 128 hidden units, deterministic; (4) **MC-Dropout**: LSTM with dropout 0.1 active at inference, 100 forward passes; (5) **Deep Ensemble**: 5 independently trained LSTMs; (6) **TimeGrad** (**?**): autoregressive RNN + DDPM decoder; (7) **CSDI** (**?**): transformer-based conditional score diffusion. All baselines use the same train/val/test splits and evaluation pipeline.

4.3 Main Results

Table **??** presents probabilistic forecasting performance at the 10-minute horizon. SolarSDE achieves the lowest CRPS across all horizons, with the improvement most pronounced at longer horizons (> 5 minutes) where cloud dynamics dominate over persistence effects. The prediction intervals achieve near-nominal 90% coverage while remaining significantly sharper (lower PINAW) than deep

Table 2: Ablation study at 10-minute horizon. Each variant removes one component.

ID	Variant	CRPS ↓	ΔCRPS (%)
A1	SolarSDE (full)	–	0.0
A2	− CTI gating	–	–
A3	− CS-VAE (PCA features)	–	–
A4	− Score Matching (linear decoder)	–	–
A5	− Neural SDE (deterministic ODE)	–	–
A6	− SDE Matching (adjoint training)	–	–
A7	− Meteorological covariates	–	–

ensembles and MC-dropout — demonstrating that CTI-conditioned diffusion produces tighter intervals during low-uncertainty periods without sacrificing coverage. On ramp events, SolarSDE's CRPS advantage is amplified, as the CTI mechanism correctly widens prediction intervals before ramps occur.

4.4 Ablation Study

Table **??** isolates the contribution of each component. Removing CTI gating (A2) degrades CRPS, confirming that state-dependent diffusion is not merely a convenience but a performance-critical design choice. Replacing the learned VAE latent space with PCA-reduced pixel features (A3) also hurts substantially, indicating that the VAE learns a more informative manifold for dynamics modeling. Replacing the score-matching decoder with a linear projection (A4) worsens calibration, consistent with the hypothesis that real irradiance distributions require a non-Gaussian generative model. Setting the diffusion to zero (A5, deterministic ODE) collapses the prediction intervals and destroys probabilistic calibration, underscoring the necessity of stochastic modeling. Switching from SDE Matching to adjoint training (A6) yields comparable accuracy but increases training time by $\sim 5\times$. Removing meteorological covariates (A7) produces a modest degradation, suggesting that the imagery captures most dynamics but meteorological features provide useful auxiliary signal.

4.5 CTI Analysis

We validate that CTI captures physically meaningful cloud dynamics through four analyses. First, CTI correlates strongly with the rolling standard deviation of observed GHI ($\rho > 0.5$, Spearman), confirming it tracks irradiance variability. Second, forecast CRPS increases monotonically across CTI quartiles: the model is most accurate under stable skies and least accurate under turbulent conditions, exactly as intended. Third, K-means clustering of CTI values into four regimes reveals distinct irradiance distributions for each: a sharp unimodal peak for the "clear" regime, a broad bimodal distribution for "broken cloud,"

and a narrow low-irradiance peak for "overcast." Fourth, t-SNE visualization of the latent space shows clear organization along physically meaningful axes (clear-sky index, cloud cover fraction, time of day).

4.6 Economic Value

We simulate a grid operator making spinning reserve commitments every 5 minutes, sizing reserves at the 95th percentile of the predictive distribution. Reserve capacity costs \$50/MWh (CAISO marginal reserve cost) and under-reserve shortfalls incur a \$1,000/MWh penalty. SolarSDE's tighter prediction intervals during low-CTI periods allow the operator to hold less reserve without increasing reliability risk. Extrapolated to a 1 GW solar plant, this translates to annual reserve cost savings compared to the next-best baseline (deep ensemble). The CTI provides the operator with an actionable signal: when CTI is low, reserve commitments can be safely reduced; when CTI spikes, reserves should be increased.

5 Discussion

CTI as an operational tool. Beyond its role in the Neural SDE, the CTI has standalone value as an operational indicator. Grid operators currently lack a single number that summarizes "how uncertain should we be right now." The CTI fills this gap. It can be displayed on control room dashboards, used as a trigger for reserve policy changes, or fed into downstream optimization algorithms. Because it is derived deterministically from the VAE's latent space, it incurs negligible computational cost.

Limitations. Several limitations merit discussion. First, we train and evaluate on a single location (Golden, Colorado). Generalization to diverse climates — tropical convection, marine stratocumulus, arid desert — requires additional validation, though the latent SDE + score decoder components (which operate on irradiance time series, not raw images) can be deployed at sites without sky cameras. Second, the CloudCV dataset's fisheye images are 1920×1080 with imperfect orientation; a production system would benefit from calibrated all-sky imagers. Third, the 10-second resolution is faster than most operational decision cycles; the architecture's value increases when paired with sub-minute grid control systems that can exploit the temporal granularity.

Future work. Three directions are promising: (1) a multi-site foundation model trained on diverse sky camera networks, where the VAE learns location-invariant cloud representations; (2) extension to satellite imagery (GOES-16/17), which would enable CTI-like indices over entire regions rather than single points; and (3) integration with battery storage optimization, where the CTI directly informs charge/discharge decisions.

6 Conclusion

We presented SolarSDE, a probabilistic solar irradiance nowcasting system that combines a Cloud-State VAE, a CTI-conditioned Latent Neural SDE, and a conditional score-matching decoder. The Cloud Turbulence Index provides a principled, interpretable mechanism for state-dependent uncertainty quantification: the model "knows" when to be confident and when to be cautious, because this behavior is architecturally encoded through the physics of cloud turbulence. Experiments demonstrate improved probabilistic calibration, sharper prediction intervals, and superior ramp event detection compared to seven baselines. The economic analysis confirms that these improvements translate to real operational value for grid operators integrating solar generation.

References

G. Bartosh, B. Lim, and J. Berner. SDE Matching. In *International Conference on Learning Representations (ICLR)*, 2025.

W. F. Holmgren, C. W. Hansen, and M. A. Mikofski. pvlib python: a python package for simulating the performance of photovoltaic energy systems. *Journal of Open Source Software*, 3(29):884, 2018.

International Energy Agency. *Renewables 2024*. IEA, Paris, 2024.

P. Ineichen and R. Perez. A new airmass independent formulation for the Linke turbidity coefficient. *Solar Energy*, 73(3):151–157, 2002.

E. B. Iversen, J. M. Morales, J. K. Møller, and H. Madsen. Probabilistic forecasts of solar irradiance using stochastic differential equations. *Environmetrics*, 25(3):152–164, 2014.

S. Jongaramrungruang, F. Marangoni, P. Szcurek, et al. Day-ahead probabilistic forecast of solar irradiance: a stochastic differential equation approach. In *IEEE PES General Meeting*, 2018.

P. Kidger, J. Foster, X. Li, and T. J. Lyons. Neural SDEs as infinite-dimensional GANs. In *International Conference on Machine Learning (ICML)*, 2021.

X. Li, T.-K. L. Wong, R. T. Q. Chen, and D. Duvenaud. Scalable gradients for stochastic differential equations. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020.

J. Perr-Sauer, R. White, and K. Munch. CloudCV 10-second sky images and irradiance dataset. NREL Data Catalog, Golden, CO, 2024.

K. Rasul, C. Seward, I. Schuster, and R. Vollgraf. Autoregressive denoising diffusion models for multivariate probabilistic time series forecasting. In *International Conference on Machine Learning (ICML)*, 2021.

Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations (ICLR)*, 2021.

Y. Sun, V. Venugopal, and A. R. Brandt. Short-term solar power forecast with deep learning: exploring optimal input and output configuration. *Solar Energy*, 188:730–741, 2019.

Y. Tashiro, J. Song, Y. Song, and S. Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

D. Yang and D. van der Meer. Post-processing in solar forecasting: Ten overarching thinking tools. *Renewable and Sustainable Energy Reviews*, 140:110735, 2021.